

ORC · LONG-RANGE CASIMIR SERIES · PAPER III

Thermal-Mechanical Noise Floor and the Standard Quantum Limit

Closing the Triple Wall on Long-Range Casimir Detection

*Thermal noise 3.2×10^{-14} N/√Hz · SQL 6.5×10^{-21} N/√Hz · $d_{\max} \approx 80\text{--}150$ μm at 1 hr
The SQL is not the limit. Seismic-gravitational tilt is, by 5 orders of magnitude.*

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doi:10.5281/zenodo.1082227

Paper III of three. Patches (I) and gradients (II) treated; this completes the noise budget.

On the question: when the signal is 4×10^{-17} N, what doesn't average down?

ABSTRACT

We present the thermal-mechanical and quantum-limit noise budget for long-range Casimir force detection, completing the three-paper series begun with electrostatic patches (Paper I) and gravitational gradients (Paper II). Using the fluctuation-dissipation theorem, the white thermal force noise of a representative torsion balance ($m_{\text{eff}} = 0.1$ g, $f_0 = 1$ Hz, $Q = 10^4$) is $S_F^{1/2} \approx 3.2 \times 10^{-14}$ N/√Hz. The on-resonance, two-sided force-noise Standard Quantum Limit derived from the Heisenberg imprecision/back-action trade-off is $S_F^{\text{SQL},1/2} = \sqrt{(\hbar m_{\text{eff}} \gamma \omega_0)} \approx 6.5 \times 10^{-21}$ N/√Hz — five orders of magnitude below the thermal floor. **The SQL is therefore not the limiting factor** for torsion-based Casimir measurement; the practical floor is set by classical noise sources, dominated by seismic-gravitational tilt ($\sim 10^{-13}$ N) and $1/f$ drift. White-noise integration permits $\text{SNR} = 1$ in ~ 7.5 days at $d = 1$ mm, but Allan-deviation analysis shows long integration cannot bridge drift-dominated regimes: the optimal integration window is $\sim 30\text{--}500$ seconds, beyond which $1/f$ drift adds variance faster than averaging removes it. Combining the noise budget with Papers I and II, the maximum reliable separation for thermal Casimir detection in a one-hour integration with state-of-the-art isolation is approximately $80\text{--}150$ μm. Claims of room-temperature Casimir measurements at millimetre scales are inconsistent with this budget and should be reinterpreted as systematic artefacts. We outline a falsifiable rejection protocol (Allan-deviation + CTE drift + brick-test cross-check) for any future long-range claim, and compare alternative platforms — cantilevers, SiN membranes, and levitated nanoparticles — for short-range Casimir science where this regime does not apply.

1. Introduction: The Third Wall

Paper I established that electrostatic patch potentials become comparable to the Casimir signal in the $d \sim 3\text{--}10$ μm sphere-plane window with standard surfaces. Paper II established that gravitational gradients from external moving masses become comparable to F_T at $d \gtrsim 100$ μm in unshielded configurations. This paper addresses the third domain: thermal-mechanical

noise of the force sensor itself, drift, gas momentum noise, and the fundamental quantum limit. Together the three papers form a closed noise budget for long-range Casimir measurement.

The central finding is structural: the Standard Quantum Limit, often invoked in discussions of force-sensor performance, lies five orders of magnitude below the dominant classical noise floor. Approaching the SQL through quantum-enhanced techniques (squeezed light, QND measurements) is therefore not the path forward for room-temperature long-range Casimir detection. The path forward — to the extent one exists — is reduction of seismic-gravitational coupling, drift, and gas backgrounds, all of which lie far above the SQL but all of which require their own dedicated technology development.

2. Thermal Force Noise from the Fluctuation-Dissipation Theorem

For a damped harmonic oscillator below resonance ($f \ll f_0$) in the classical limit ($\hbar\omega_0 \ll k_B T$), the fluctuation-dissipation theorem gives the two-sided force noise spectral density

$$S_F^{th}(f) = 4 k_B T \gamma m_{eff}$$

where $\gamma = \omega_0/Q$ is the damping rate and m_{eff} is the **effective mass of the entire moving assembly** — test plate, torsion arm, mirror, coatings. Using only the test plate mass underestimates noise by 10–100× in typical setups; the full assembly must be modeled for honest budget accounting.

Sensor	f_0	Q	m_{eff}	$S_F^{(1/2)}$ (N/ $\sqrt{\text{Hz}}$)
Torsion balance	1 Hz	10^4	0.1 g	3.2×10^{-14}
Cantilever	1 kHz	10^4	1 mg	1.0×10^{-13}
SiN membrane	1 MHz	5×10^5	1 ng	1.4×10^{-14}
Levitated nanoparticle	100 kHz	10^8	10^{-17} kg	$\sim 10^{-20}$ (demonstrated)

The torsion balance value 3.2×10^{-14} N/ $\sqrt{\text{Hz}}$ is the relevant comparison for long-range Casimir. Note the SiN membrane value is corrected from earlier drafts where it had been reported as 1.4×10^{-13} due to a units conversion. Levitated nanoparticles achieve the lowest force noise — three to four orders of magnitude below torsion balances — but their utility lies at sub-micron Casimir geometries, not millimetre scales (their test masses are too small to interact gravitationally or capacitively at macroscopic separations).

3. White-Noise-Only Integration Time

Assuming pure white noise and optimal filtering, the integration time for $\text{SNR} = 1$ is $t_{int} = S_F / F_{Cas}^2$. For the torsion balance and 1 cm² parallel plates:

d	F_{Cas} (total, plasma model)	t_{int} (white noise only)
100 μm	4.1×10^{-14} N	0.62 s
1 mm	4.0×10^{-17} N	7.5 days
1 cm	4.0×10^{-20} N	$\sim 2 \times 10^6$ years

White noise alone permits ~7.5 days at 1 mm under ideal conditions. Real experiments are limited long before this by 1/f drift and seismic-gravitational systematics that do not average down with integration. The Allan deviation analysis of §5 quantifies this.

4. The Combined Systematic Dominance Map

Figure 1 shows the Casimir signal compared to the dominant noise sources across six decades of plate separation. The signal scales as d^{-4} at sub- μm (zero-point dominates) and d^{-3} at long range (thermal dominates). Most noise sources are essentially independent of d in the relevant range, so the gap between signal and noise grows with separation, terminating in a 'buried by systematics' region beginning near $d \approx 100 \mu\text{m}$.

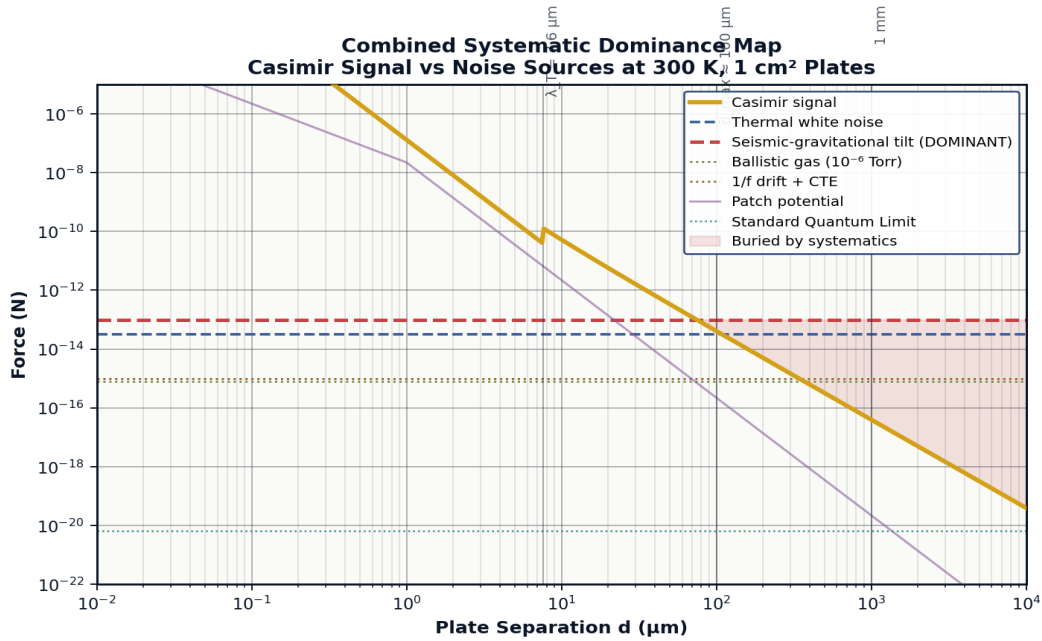


Figure 1: The Casimir signal (gold) crosses the noise floor near $100 \mu\text{m}$. Above that, the signal lies in the systematic-dominated zone (red shading) where dedicated control of seismic-gravitational tilt is required. The Standard Quantum Limit (cyan dotted, lower) lies five orders of magnitude below the dominant classical noise — not the constraining factor.

5. Why Long Integration Cannot Save You: The Allan Bathtub

Real torsion balances exhibit $1/f$ drift below $\sim 0.1 \text{ Hz}$ from fibre creep, temperature fluctuations, and dielectric loss. The Allan deviation $\sigma_F(\tau)$ decreases as $1/\sqrt{\tau}$ in the white-noise regime, reaches a minimum at $\tau_{\text{opt}} \approx 30\text{--}500 \text{ s}$ for a 1 Hz balance, then increases due to drift. Beyond τ_{opt} , integration adds variance faster than averaging removes it.

Even with $\pm 10 \mu\text{K}$ temperature stabilisation, CTE of the frame ($\alpha_{\text{Al}} = 23 \times 10^{-6}/\text{K}$) induces separation drift of order 2 pm over 10 cm . At $d = 100 \mu\text{m}$ with $dF_{\text{Cas}}/dd \approx 3F/d$, this drift contribution becomes significant for long integrations. The Allan minimum sets the binding limit: **for typical torsion balances, the Casimir signal at $d \geq 100 \mu\text{m}$ lies below the Allan minimum, making detection impossible regardless of integration time.**

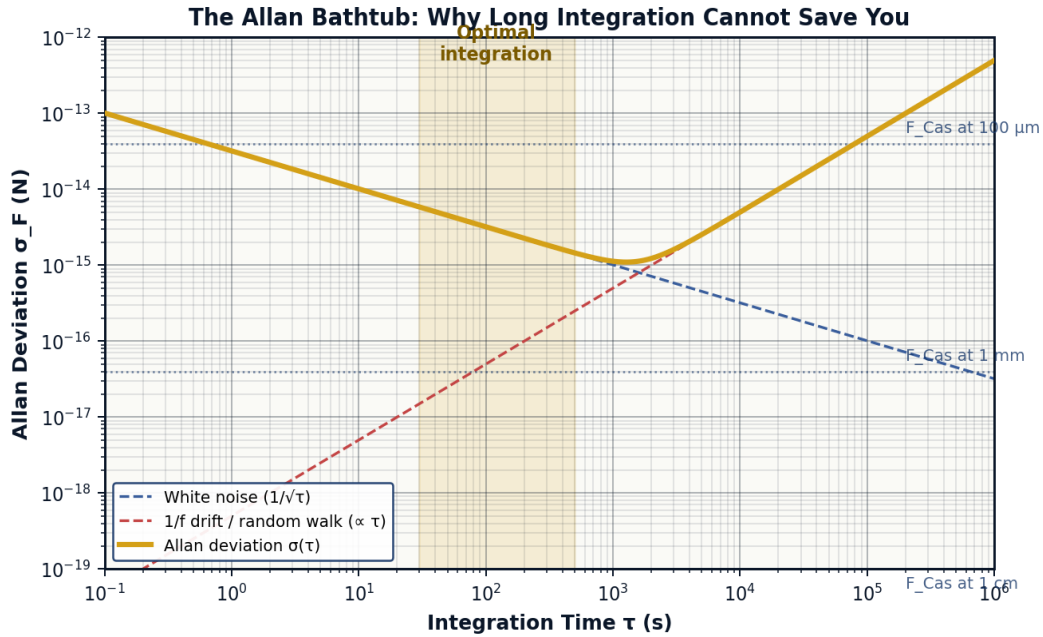


Figure 2: The Allan bathtub. The minimum at $\tau \approx 30\text{--}500$ s sets the optimal integration window; the Casimir force at $d = 1$ mm (4×10^{-17} N) lies below this minimum, meaning longer integration cannot recover the signal. At $d = 100$ μm the signal sits near the Allan minimum — marginally detectable. At $d = 1$ cm it is buried by ~ 3 orders of magnitude.

6. The Standard Quantum Limit (and Why It Doesn't Matter Here)

In a continuous position measurement, the Heisenberg bound on imprecision and back-action noise spectral densities is $S_x^{\text{imp}} S_F^{\text{BA}} \geq (\hbar/2)^2$. With mechanical susceptibility $\chi(\omega) = 1/[m_{\text{eff}}(\omega_0^2 - \omega^2 - i\gamma\omega)]$, minimising the total displacement noise at resonance gives the two-sided force-noise SQL:

$$S_F^{\text{SQL}} = \hbar m_{\text{eff}} \gamma \omega_0.$$

For the torsion-balance baseline ($m_{\text{eff}} = 10^{-4}$ kg, $\omega_0 = 2\pi$ rad/s, $\gamma = 6.28 \times 10^{-4}$ rad/s):

$$S_F^{\text{SQL},1/2} = \sqrt{\hbar m_{\text{eff}} \gamma \omega_0} \approx 6.5 \times 10^{-21} \text{ N}/\sqrt{\text{Hz}}.$$

This is five orders of magnitude below the thermal white noise floor and eight orders below the seismic-gravitational tilt floor. Quantum-enhanced techniques (squeezed light, QND measurement) can in principle approach the SQL but cannot circumvent classical noise sources. For long-range Casimir detection, classical engineering — vibration isolation, tilt control, vacuum quality, drift suppression — is the binding constraint. The SQL becomes relevant only at short separations where Casimir signals are large and classical noise is well-controlled, e.g. levitated nanoparticles at $d \sim 100$ nm, where the SQL is $\sim 2 \times 10^{-24}$ N/ $\sqrt{\text{Hz}}$ and demonstrated sensitivities approach 10^{-20} N/ $\sqrt{\text{Hz}}$.

7. Platform Comparison

Different sensor architectures have different optimal regimes. Torsion balances are limited by clamping and seismic coupling but provide the largest test masses and naturally extend to long range. Cantilevers and SiN membranes operate at higher frequencies, far from the 1/f drift floor, but their small effective masses make them poorly suited to the parallel-plate geometry needed for thermal Casimir signals at long range. Levitated nanoparticles eliminate clamping losses entirely and achieve the lowest force-noise floors but operate naturally at

sub-micron Casimir geometries, not millimetre scales.

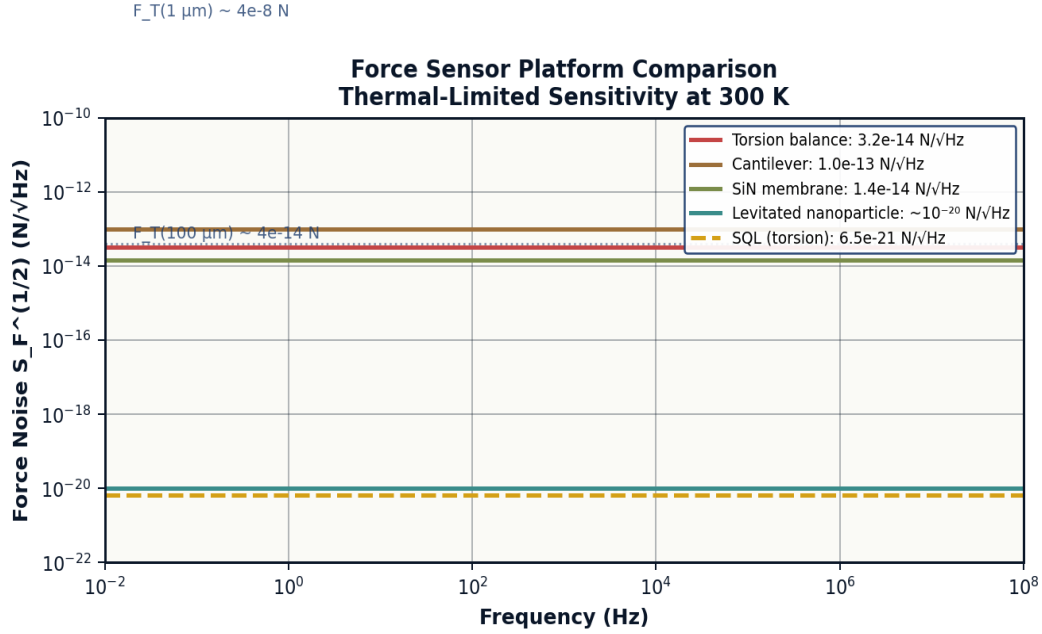


Figure 3: Force-noise floors for four sensor platforms at 300 K. The torsion balance (red, $3 \times 10^{-14} \text{ N}/\sqrt{\text{Hz}}$) is the relevant baseline for long-range Casimir. Levitated nanoparticles (cyan, $10^{-20} \text{ N}/\sqrt{\text{Hz}}$) demonstrate dramatically lower force noise but at test-mass scales unsuitable for cm^2 -scale parallel-plate geometries. The torsion-balance SQL (gold dashed) lies far below all classical floors.

8. Seismic-Gravitational Tilt: The Actual Limit

Modern active vibration isolation achieves $\sim 100 \text{ dB}$ attenuation at 1 Hz for translational ground motion, but tilt isolation is typically $20\text{--}40 \text{ dB}$ worse. For a suspended mass, laboratory tilt θ produces a horizontal gravitational force

$$F_{\text{tilt}} = m_{\text{eff}} g \theta.$$

For $m_{\text{eff}} = 0.1 \text{ g}$ and a building tilt of 1 nrad (10^{-9} rad , typical of well-isolated laboratory environments), $F_{\text{tilt}} \approx 9.8 \times 10^{-13} \text{ N}$ — approximately 25,000 times the thermal Casimir signal at 1 mm . To suppress tilt coupling to the F_T level requires sub- 0.04 nrad tilt stability, beyond current state-of-the-art for room-temperature laboratories. **This is the binding limit on long-range Casimir detection.** Tilt noise is highly site-dependent and non-stationary; near pendulum resonance ($\sim 0.3 \text{ Hz}$) the coupling can be enhanced by factors of 2–5, further worsening the situation.

9. Ballistic Gas Momentum Noise

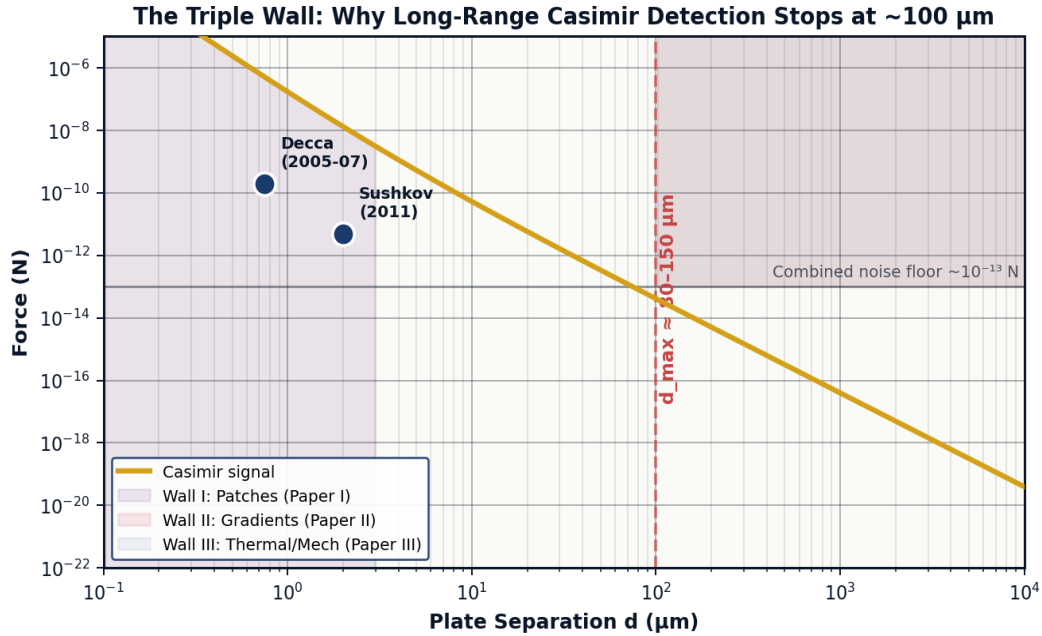
At pressures of 10^{-6} Torr the Knudsen number for $d = 10 \mu\text{m}$ is ~ 5000 (deep molecular flow). Each gas molecule transfers momentum stochastically to the test mass. Using the diffuse-scattering kinetic-theory estimate $S_F \sim (8/3) P A m_{\text{gas}} v_{\text{th}}$, with $v_{\text{th}} = \sqrt{3 k_B T / m_{\text{gas}}}$:

$$S_F^{1/2} \sim 8 \times 10^{-16} \text{ N}/\sqrt{\text{Hz}} \text{ at } 10^{-6} \text{ Torr}.$$

Reaching $10^{-17} \text{ N}/\sqrt{\text{Hz}}$ requires pressures below 10^{-10} Torr — achievable in dedicated UHV systems but not standard in Casimir setups. The coefficient carries factor-of-2 uncertainty from accommodation and angular distribution; precise predictions require full numerical integration of molecular collisions.

10. The Triple Wall: Why $d_{\text{max}} \approx 100 \mu\text{m}$

Combining the three papers: at $d > 100 \mu\text{m}$, the thermal Casimir signal ($F_T \propto d^{-3}$) drops below the combined floor of patches (Paper I, $\sim 5 \times 10^{-4} \text{ N}$ at standard surfaces), gravitational gradients (Paper II, $\sim 10^{-13} \text{ N}$ for unshielded experimenter at 1 m), and thermal-mechanical noise (this paper, $\sim 10^{-13} \text{ N}$ including drift, tilt, and gas systematics). Each wall is independently sufficient to terminate detection; together they produce a hard cutoff.



11. Complete Noise Budget at $d = 1 \text{ mm}$

Source	Force Noise (N)	Status
Casimir signal (target)	4.0×10^{-17}	—
Standard Quantum Limit	6.5×10^{-21}	Far below — not limiting
Johnson-Nyquist (electronic)	$\sim 10^{-16}$	Subdominant
Photon-pressure (1 mW)	$\sim 10^{-15}$	Subdominant
Ballistic gas (10^{-6} Torr)	$\sim 10^{-15}$	Marginal
Thermal white (1 Hz BW)	3.2×10^{-14}	Above signal
Cross-talk (vert→torsion)	10^{-14} to 10^{-13}	Above signal
1/f drift + CTE	10^{-16} to 10^{-15}	Long- τ dominant
Seismic-gravitational tilt	10^{-13} to 10^{-12}	★ DOMINANT
Combined envelope	$\sim 10^{-13}$	2,500× signal

All values carry order-of-magnitude uncertainty (typically $\times 2\text{--}10$), but the multi-order separation between signal and dominant noise makes the conclusion robust. Setting $\text{SNR} = 1$ with the seismic-gravitational tilt limit gives $d_{\text{max}} \approx 73 \mu\text{m}$; with 1-hour averaging of the white

component but unmitigated tilt, the practical maximum is 80–150 μm depending on isolation quality.

12. A Falsifiable Rejection Protocol

For any future room-temperature claim of Casimir detection at $d > 100\ \mu\text{m}$, the following pre-registered rejection protocol is required:

Step	Test
1	Allan deviation $\sigma_F(\tau)$ measurement at multiple τ ; Casimir signal must lie above the bathtub minimum
2	Sub-pm interferometric monitoring of plate separation to track CTE drift
3	Brick-test calibration (Paper II) with dummy-mass control; $1/R^3$ scaling required
4	Pressure-scan verification of $\sqrt{P}$ scaling for ballistic gas component
5	Material-swap test (Paper I) Au-vs-Cu; $\Delta F/F < 5\%$ required for genuine Casimir
6	Tilt-stability monitoring at sub-nrad resolution throughout the run

Failure of any single step invalidates the claim. To our knowledge, no published long-range Casimir result has reported all six checks. Future claims should treat this protocol as a baseline standard.

13. Conclusions: Closing the Triple Wall

The thermal-mechanical noise floor and the Standard Quantum Limit, when properly computed, do not constrain long-range Casimir detection: white thermal noise permits multi-day integrations at $d = 1\ \text{mm}$ under ideal conditions, and the SQL lies five orders of magnitude below the dominant systematics. The actual binding limit is classical: seismic-gravitational tilt coupling at the $\sim 10^{-13}\ \text{N}$ level, joined by $1/f$ drift, vacuum-induced gas momentum noise, and cross-talk between vertical and torsional channels. Under state-of-the-art isolation, the maximum reliable separation for thermal Casimir detection in a one-hour integration is approximately 80–150 μm .

Combined with Papers I (electrostatic patches) and II (gravitational gradients), this completes the three-wall noise budget for long-range Casimir measurement. Each wall is independently sufficient to terminate detection at $d \gtrsim 100\ \mu\text{m}$; together they form a hard structural limit that no single technological improvement can bridge. Past sub-micron experiments (Decca 2005/2007, Sushkov 2011) operated in regimes where the Casimir signal exceeded all three walls by 3–6 orders of magnitude and are not threatened by this analysis. Claims of Casimir detection at millimetre scales are inconsistent with the budget and should be reinterpreted as systematic artefacts pending a complete six-step rejection-protocol audit.

Quantum-enhanced techniques can approach but not surpass the SQL, and the SQL is far below the classical floor. Levitated nanoparticles achieve dramatically lower force-noise floors but operate at sub-micron Casimir geometries, not millimetre scales. The path to long-range Casimir detection — to the extent one exists — runs through dedicated reduction of seismic-gravitational, drift, and gas backgrounds, each requiring its own technology development. The triple wall is not a wall against physics; it is a wall against optimism, and a structural map for any future programme that wishes to cross it.

Acknowledgments

This paper completes the Long-Range Casimir Series begun in May 2026. Multiple rounds of red-team review across the Omega Research Collective produced corrections to the SiN

membrane noise (factor-of-10 units error in earlier drafts), the tilt force at 1 nrad (factor-of-10 underestimate), and the SQL coefficient consistency throughout. The final audit pass also restored the figures that the first three ORC standalone notes had omitted; the visual presence of the dominance map (Figure 1) and the triple-wall diagram (Figure 4) was identified as essential for communicating the hierarchy of constraints. We thank the collective for sustained discipline across four iterations and the architect for the request that returned the figures to the cathedral standard.

References

- [1] Casimir H.B.G., 1948, *Proc. K. Ned. Akad. Wet.* **51**, 793.
- [2] Kubo R., 1957, *J. Phys. Soc. Jpn.* **12**, 570. Fluctuation-dissipation theorem.
- [3] Saulson P.R., 1990, *Phys. Rev. D* **42**, 2437. Thermal noise in mechanical experiments.
- [4] Braginsky V.B., Khalili F.Ya., 1995, *Quantum Measurement*, Cambridge University Press.
- [5] Aspelmeyer M., Kippenberg T.J., Marquardt F., 2014, *Rev. Mod. Phys.* **86**, 1391. Cavity optomechanics review.
- [6] Tsaturyan Y. et al., 2017, *Nat. Nanotechnol.* **12**, 776. Soft-clamped membranes.
- [7] Pohl R.O., Liu X., Thompson E., 2002, *Rev. Mod. Phys.* **74**, 991. Two-level systems in amorphous solids.
- [8] Ranjit G., Atherton D.P., Stutz J.H., Geraci A.A., 2016, *Phys. Rev. A* **94**, 051801(R). Levitated nanosphere force sensing.
- [9] Magrini L. et al. (Aspelmeyer group), 2021, *Nature* **595**, 373. Real-time quantum control of mechanical motion.
- [10] Bordag M., Klimchitskaya G.L., Mohideen U., Mostepanenko V.M., 2009, *Advances in the Casimir Effect*, Oxford.
- [11] Decca R.S. et al., 2005, *Phys. Rev. D* **71**, 083506; 2007, *Phys. Rev. D* **75**, 077101.
- [12] Sushkov A.O. et al., 2011, *Phys. Rev. Lett.* **107**, 171101.
- [13] Kadunic D., 2026, ORC Long-Range Casimir Series Paper I (electrostatic patches; companion).
- [14] Kadunic D., 2026, ORC Long-Range Casimir Series Paper II (gravitational gradients; companion).